5.7.1 Atomic Theory

After completing this chapter you should be able to relate properties of elements, compounds and mixtures to scientific models, theories and laws.

For Questions 1 to 10 choose the best answer (A), (B), (C) or (D).

1. The property which distinguishes one element from another is:

(A) The number of protons in the nuclei of the atoms.
(B) The number of neutrons in the nuclei of the atoms.
(C) The number of electrons in orbit around the nuclei.
(D) The mass of their atoms.

2. The table shows the number of protons, neutrons and electrons in atoms of W, X, Y and Z.

	Number of protons	Number of neutrons	Number of electrons
W	15	16	15
X	16	16	16
Y	17	17	16
Z	16	17	18

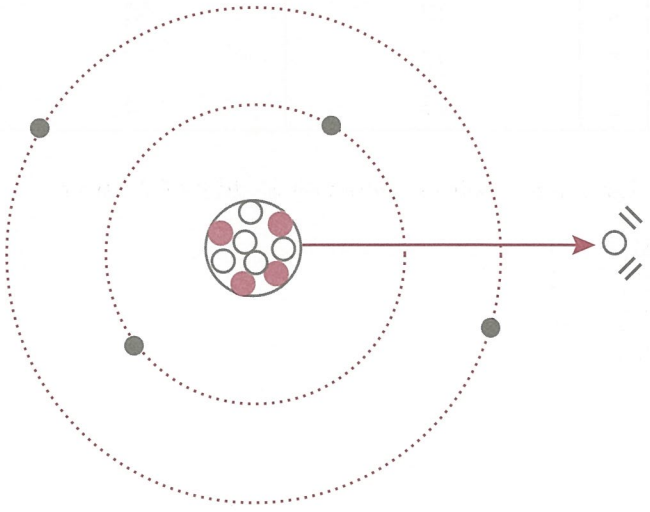
Which atoms are atoms of the same element?

(A) W and X.
(B) X and Y.
(C) X and Z.
(D) Y and Z.

3. The diagram represents a neutral atom. It is releasing a subatomic particle.

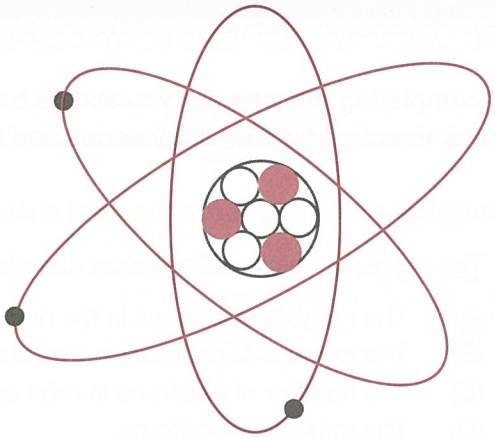
What type of particle is being released?

(A) Electron.
(B) Neutron.
(C) Nucleus.
(D) Proton.



Use the following information to answer the next TWO questions.

The diagram represents a model of an element.



4. Which statement about this atom is correct?

(A) It contains 3 protons, 3 electrons and 4 neutrons.
(B) It contains 4 protons, 3 electrons and 3 neutrons.
(C) It contains 3 protons, 4 electrons and 3 neutrons.
(D) It contains 3 protons, 4 electrons and 4 neutrons.

5. Refer to the Periodic Table at the back of this book.
Which element does this atom represent?

(A) Beryllium.
(B) Boron.
(C) Helium.
(D) Lithium.

6. Which choice correctly identifies the charge on the particles?

	Protons	Neutrons	Electrons
(A)	Positive	Negative	Negative
(B)	Positive	Neutral	Negative
(C)	Negative	Positive	Neutral
(D)	Negative	Neutral	Positive

7. The table shows information about four elements.

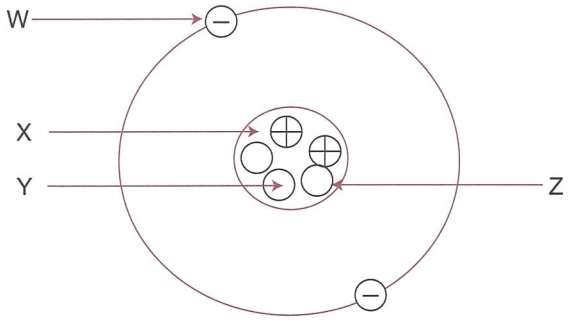
	Number of protons	Number of neutrons	Number of electrons
W	11	12	11
X	12	12	12
Y	12	13	?
Z	13	14	13

How many electrons would each atom of Y have?

(A) 11
(B) 12
(C) 13
(D) 14

Use the following information to answer the next TWO questions.

The diagram shows a model of an atom.



8. Which labelled part of the atom is the nucleus?

(A) W
(B) X
(C) Y
(D) Z

9. Given the following additional information:

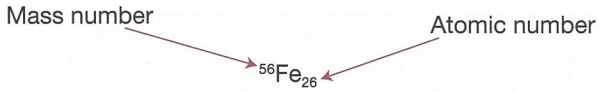
Mass number of an atom = number of protons + number of neutrons

What is the mass number of the atom shown?

(A) 3
(B) 4
(C) 5
(D) 7

10. This question refers to the following information:

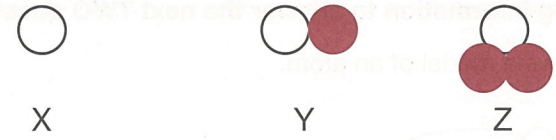
The atomic number of an element is equal to the number of protons in one atom of the element.
The mass number is the number of protons plus the number of neutrons in an atom of the element.
For any element, for example iron, this information can be written like this:



An atom of silver (Ag) has 47 protons and 61 neutrons. Which formula is correct?

(A) $^{61}Ag_{47}$
(B) $^{61}Ag_{108}$
(C) $^{108}Ag_{61}$
(D) $^{108}Ag_{47}$

11. The diagrams below represent models of the nuclei of three atoms:



X Y Z

(a) Identify the particles in these nuclei and justify your names.

○ =

Proton

● =

Neutron

(b) Identify the element to which each nucleus belongs and justify your choices.

X =

Hydrogen - 1 proton and no neutrons

Y =

Hydrogen - 1 proton and 1 neutron (deuterium)

Z =

Hydrogen - 1 proton and 2 neutrons (tritium)

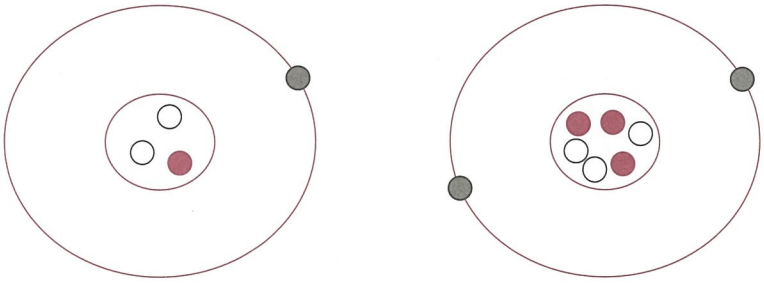
12. Complete the table to summarise the properties of the particles that make up atoms.

Particle	Location	Charge
Neutron	In nucleus	Neutral
proton	In nucleus	Positive
Electron	orbiting nucleus	Negative

13. Atoms of elements are always neutral because they contain equal numbers of protons and electrons. What would be the charge remaining on the particle formed if an atom lost an electron? Explain your answer.

Positive charge because it would have one more proton than it has electrons.

14. A student drew diagrams to represent two different atoms:



Atom X Atom Y

(a) What evidence is in the diagrams to indicate that these are atoms of *different* elements?

There are different numbers of protons in the nucleus.

(b) Indicate an error in the student's diagrams and state how to correct it.

Element 4 should have 3 electrons to equal the 3 protons; or 2 protons to equal the 4 electrons.

15. Describe a model that has been developed to describe atomic structure.

Bohr model.
Central nucleus containing protons and neutrons.
Protons positively charged.
Neutrons neutral.
Around the nucleus in specific energy levels are electrons.
Electrons are negatively charged.
The number of protons equals the number of electrons.